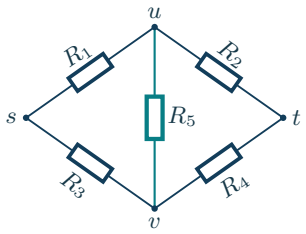
THE WHEATSTONE BRIDGE AND RANDOM WALKS

A bridge between circuits and chance

Kirchhoff's averaging rule links voltages, probabilities, expected journeys and squared rectangles.

1 The A-level bridge



$$V_s = V, \qquad V_t = 0.$$

$R_5 > 0$ is the finite-resistance detector branch. The random walk will use the five resistor edges.

2 Balance means equal potentials

At balance, $V_u = V_v$, so the middle branch carries zero current. The two arms are voltage dividers:

$$V_u = V\frac{R_2}{R_1+R_2}, \qquad V_v = V\frac{R_4}{R_3+R_4}.$$

With the labels shown,

$$\boxed{V_u = V_v \iff \frac{R_1}{R_2} = \frac{R_3}{R_4}}.$$

The effective resistance is V/I_{total}. At balance,

$$R_{\text{eff}}(s,t) = \left[\frac{1}{R_1+R_2} + \frac{1}{R_3+R_4}\right]^{-1}.$$

3 A voltage becomes a hitting probability

On a finite, connected, undirected resistor network, choose the next edge with probability proportional to its conductance:

$$c_{xy} = \frac{1}{R_{xy}}, \qquad p_{xy} = \frac{c_{xy}}{\sum_z c_{xz}}, \qquad \underbrace{V_x = \sum_y p_{xy}V_y}_{\text{Kirchhoff's law at an interior node}}.$$

Let q_x be the probability, starting at x, of reaching s before t. Looking at the first step gives

$$q_x = \sum_y p_{xy}q_y, \qquad q_s = 1, \quad q_t = 0 \implies \boxed{q_x = \frac{V_x}{V}}.$$

The same averaging rule and boundary values give the same solution. Equal resistors make all incident edges equally likely.

4 Resistance measures the expected round trip

Let h_{st} be the expected number of steps to first reach t starting at s. The **commute time** is

$$\boxed{K_{st} = h_{st} + h_{ts} = 2\left(\sum_{e\in E} c_e\right) R_{\text{eff}}(s,t)}.$$

Count each undirected resistor edge once in the sum. For m equal resistors of resistance R_0, $\boxed{K_{st} = 2m\,\dfrac{R_{\text{eff}}(s,t)}{R_0}}$. Time is measured in steps along edges.

5 Five equal resistors

Set $R_1 = \cdots = R_5 = R_0$. Symmetry gives

$$V_u = V_v = \frac{V}{2}, \qquad I_5 = 0.$$

The two outer paths each have resistance $2R_0$:

$$R_{\text{eff}}(s,t) = (2R_0) \parallel (2R_0) = R_0.$$

There are still five edges in the walk:

$$\boxed{K_{st} = 2 \cdot 5 \cdot \frac{R_0}{R_0} = 10 \text{ steps}}.$$

Check by taking the next step

Let h_x be the expected number of steps from x to t. Then $h_t = 0$. By symmetry, write $h_u = h_v = h$.

$$h_s = 1 + h, \qquad h = 1 + \frac{h_s + h + 0}{3}.$$

The initial 1 counts the step just taken. Solving gives

$$h = 4, \qquad h_{st} = h_s = 5.$$

The return journey has the same mean by symmetry:

$$\boxed{K_{st} = 5 + 5 = 10}.$$

A balanced edge still carries walkers

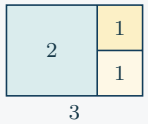
Middle edge	m	R_{eff}/R_0	K_{st}
Present	5	1	10
Removed	4	1	8

At u, the walker chooses v with probability $1/3$, even though the balanced electrical current through R_5 is zero.

The extra edge introduces detours. It also changes the total-conductance factor in the commute-time formula.

6 The rectangle becomes a walk

Three square tiles give three resistors R_0, with $R_{\text{eff}}/R_0 = H/W$.



$$\frac{R_{\text{eff}}}{R_0} = \frac{2}{3}, \qquad m = 3.$$

$$\boxed{K = 2 \cdot 3 \cdot \frac{2}{3} = 4 \text{ steps}}.$$

Tile height encodes voltage drop; tile width current in scaled units. The same Kirchhoff network now describes chance.

Sources: Doyle–Snell; Chandra et al. (1996); Aldous–Fill, §3.3; Cannon–Floyd–Parry, §1. Companion: *Kirchhoff Circuits & Squared Rectangles*.